

WASP-12b

R-0154

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_250220 · created 2026-07-16T14:30:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460726.7942 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.045
Transit depth (Rp/Rs)^2	2.28 +/- 1.35 [%]
Orbital Inclination (inc)	86.65 +/- 2.92
Airmass coefficient 1 (a1)	0.51 +/- 0.29
Airmass coefficient 2 (a2)	0.21 +/- 0.14
Scatter in the residuals of the lightcurve fit is	60.18 %
Best Comparison Star	None
Optimal Aperture	4.45
Optimal Annulus	7.05
Transit Duration (day)	0.1039 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.045 vs 0.1178 (deviation 0.61σ)
Duration fit vs published	0.1039 d vs 0.125 d (deviation 4.96σ)
Mid-time O–C	1.5 min
Depth SNR	2.8
Residual scatter	60.18 %

Light curve

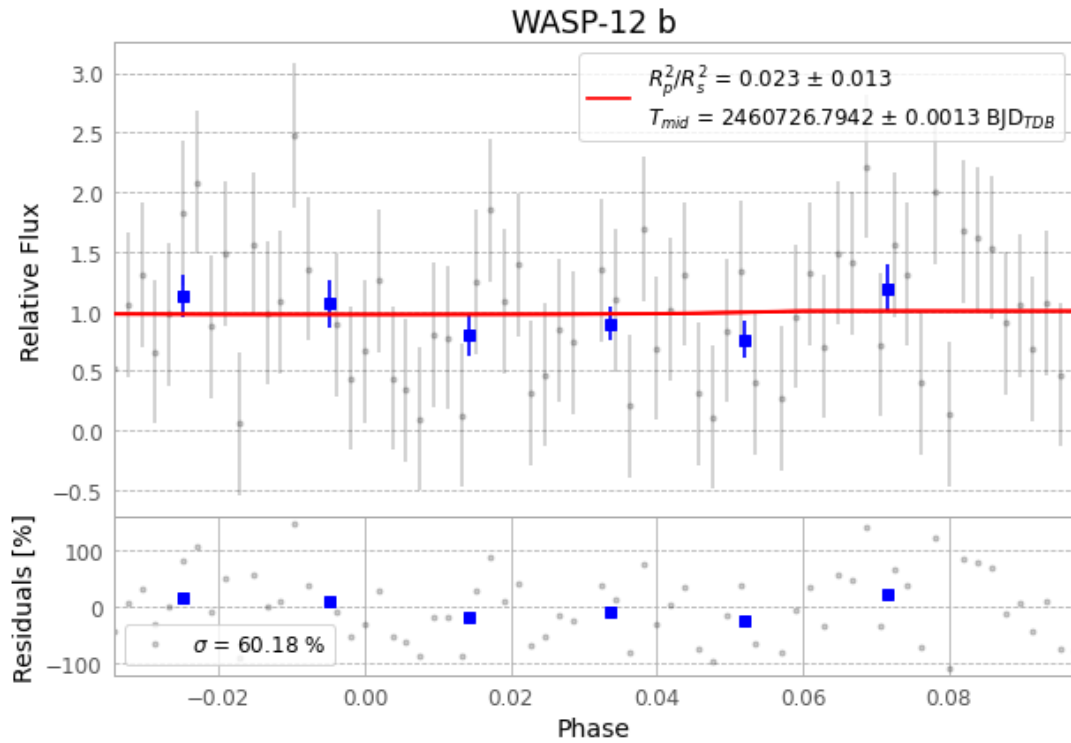


Figure 1 — FinalLightCurve_WASP-12 b_2025-02-19.png (SHA-256 122cca2e6ed433bc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 393], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5fd72f300d503efe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 6eeacf9

Data provenance

70 FITS frames (46 MB), WASP-12250220060317.FITS → WASP-12250220093015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250220.log (aa18099ce95c...), FinalLightCurve_WASP-12 b_2025-02-19.png (122cca2e6ed4...), FinalLightCurve_WASP-12 b_2025-02-19.csv (daa56a819479...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 14:30Z.